Lakshay Sangwan

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WORK EXPERIENCE

DIGII (COLLPOLL) | Product Engineer 2 — Remote (Bengaluru, Karnataka) | Nov 2023 – Present

- Built a **faculty-recruitment platform** end-to-end as the company's first **Java 25 / Spring Boot 4** microservice with a public candidate portal – ~**230 endpoints,** ~**2,500 tests**, shipped to QA in ~**4 weeks** using a multi-agent AI workflow.
- Built a **staff-appraisal engine** with rule-based rosters, multi-reviewer scoring and **pluggable self-scoring KPIs** from attendance, leave and exam data; ~**975 unit tests**.
- Architected a **multi-tenant federation hub** using **union-find transitive closure** and HMAC-signed JWTs for cross-tenant account linking.
- Designed **selfie-based attendance** with server-side geofencing, multi-layer timezone resolution and anti-spoofing checks.
- Implemented **FF1 format-preserving encryption** for platform-wide ID obfuscation with no schema changes.
- Led **AWS cost optimization**: audited 15+ ElastiCache clusters, upgraded Redis 6.2.6 to 7.1.0, right-sized instances and pruned low-traffic CloudFront distributions – **$3,400+/year saved**.
- Owned **Staff Attendance** end-to-end, replacing day-old Athena batch reports with real-time ElasticSearch sync.
- Built org-wide **reusable components**: global exception-handling advices, test frameworks for services and stored procedures, and obfuscation utilities.
- Shipped **Digital Student Passport** (QR verification) and **Support Help Centre**; perform **cross-level code reviews** and author HLDs.

AUTHBRIDGE | Software Engineer — Gurgaon, Haryana | Jun 2021 – Nov 2023

- Designed an **automated data inflow system**: zero developer intervention, **5x** scraping throughput, and compute moved from enterprise servers to consumer-grade hardware.
- Built core **PAN and Aadhaar verification** microservices behind the flagship background-verification product.
- Developed internal tools for data extraction, validation pipelines and batch processing.

PERSONAL PROJECTS

DOCUKIT — Cloudflare Pages, React (AI-assisted), PDF.js

Privacy-first document toolkit for PDF/image processing – merge, split, compress, convert – all processed **client-side with zero server uploads**. Owned product direction, security design, and SEO optimization.

HABITFLOW — Cloudflare Workers, D1, Hono, React (AI-assisted)

Serverless habit tracking app on **Cloudflare edge stack**. Designed architecture, database schema, and API contracts; frontend built with AI assistance.

EDUCATION

B.Tech. in Computer Science Engineering — Kurukshetra, Haryana | May 2021

UIET, Kurukshetra University

SKILLS

Languages: Java, JavaScript, TypeScript, SQL, HTML/CSS
Backend: Spring Boot (up to 4.x), Hibernate (JPA), REST APIs, Microservices
Frontend: AngularJS, React (familiar)
Databases: MySQL, PostgreSQL, Redis/ElastiCache, ElasticSearch
Cloud & Infra: AWS (EC2, ECS, SQS, Athena, CloudFront, ALB, ElastiCache), Cloudflare (Workers, Pages, D1)
Tools: Git, Linux, Bash scripting, Playwright, GitHub Actions